

DD-Mamba: Dual-Domain State-Space Modeling with Zero-Initialized Linear Anchors for Long-Term Time Series Forecasting

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Abstract. Long-term multivariate time series forecasting must reconcile two complementary views of a signal: local, trend-driven dynamics that live in the time domain, and global periodic structure that is most compact in the frequency domain. Existing state-space (Mamba) forecasters model one view well but not both, and deep forecasters in general discard the simple linear maps that remain near state-of-the-art on several benchmarks. We present DD-Mamba, a dual-domain architecture in which a time branch (trend–seasonal decomposition with a selective state-space encoder) and a frequency branch (learnable complex spectral filtering) each produce their own forecast, fused by a learned convex gate so that no prediction can bypass either domain. Two design principles drive its accuracy: (i) *linear anchoring* — each branch embeds the strongest linear forecaster of its domain (DLinear-style projections in time, FITS-style spectral interpolation in frequency) and zero-initializes its deep heads, so training starts from an exact linear model and learns nonlinear corrections; and (ii) *capacity placement* — a bidirectional Mamba over the *variate* dimension carries the model on high-channel data, while it is disabled on small, few-channel data where it only adds overfitting capacity. On nine standard benchmarks under the look-back-96 protocol, DD-Mamba attains the best average MSE on seven datasets and matches the strongest state-space baseline on one more, improving over S-Mamba by up to 3.5% on ETTh1 at a fraction of the parameters on small datasets. Ablations confirm each design choice and surface a cautionary finding: instance normalization *helps* traffic yet *hurts* solar, so normalization must be validated per dataset.

Keywords: Time series forecasting · State-space models · Mamba · Frequency domain · Multivariate analysis

1 Introduction

Long-term forecasting of multivariate time series underpins applications from power-grid load planning to traffic management and weather-dependent operations. The field has cycled through architectural paradigms at a remarkable pace: Transformer variants with efficient attention [1–3], patch-based attention [4], inverted variate-token attention [5], and, most recently, selective state-space

models (SSMs) built on Mamba [11, 12], which offer linear-time sequence modeling with input-dependent dynamics.

Two empirical facts complicate this progress. First, *simple linear maps remain stubbornly competitive*: a per-component linear projection from the look-back window to the horizon (DLinear [6]) and a tiny complex-linear interpolation of the low-passed spectrum (FITS [10]) match or beat far larger models on several benchmarks, especially the ETT family. Deep architectures that ignore this fact spend their early training re-discovering a linear map, and on small datasets often never do. Second, *the useful inductive bias differs by domain*: local trends and regime changes are naturally expressed in the time domain, while multi-scale seasonality is compact in the frequency domain. Models committed to a single view — attention or SSMs over time steps only, or spectral mixing only [9] — systematically underuse the other.

We address both issues with DD-Mamba, a dual-domain forecaster organized around three ideas.

Dual branches that cannot be bypassed. A time branch and a frequency branch each output a complete forecast; the model’s prediction is a learned *convex combination* of the two, with the gate conditioned on both branches’ features. Because the output is constrained elementwise between the branch forecasts, no shortcut path around either domain exists — a property we verify by construction and exploit for per-sample attribution of the two domains.

Linear anchors with zero-initialized corrections. Each branch embeds the strongest *linear* forecaster of its domain as an internal backbone: seasonal/trend window-to-horizon projections [6] in the time branch, and a FITS-style complex-linear spectral upsampling [10] in the frequency branch. All deep heads are zero-initialized, so at initialization DD-Mamba *is* an exact linear model, and gradient descent learns nonlinear corrections only where the data supports them. This one choice stabilizes small-data training and removes the mean-reversion failure mode of pooled deep forecasters.

Mamba where it pays. A causal Mamba encoder models the time axis; a *bidirectional* Mamba (frequency bins and channels have no arrow of time) serves two optional roles: spectral encoding, and cross-channel mixing over the variate dimension in the spirit of S-Mamba [12]. Crucially, capacity is *placed* per dataset: the variate mixer is the dominant component on high-channel data (electricity, traffic, solar) and is disabled on few-channel small data (ETT, exchange) where ablations show it only adds overfitting capacity.

Our contributions are:

1. A dual-domain forecasting architecture whose output is a gated convex combination of per-branch forecasts, guaranteeing that both the time and the frequency view contribute to every prediction (Sect. 3).
2. A linear-anchoring scheme — domain-appropriate linear backbones with zero-initialized deep corrections — that makes the model an exact linear forecaster at initialization and yields large gains on small benchmarks (Sect. 3.2, ablated in Sect. 4.3).

3. A capacity-placement recipe for selective SSMS: causal Mamba over time where temporal microstructure matters, bidirectional Mamba over variates where cross-channel structure dominates, and neither where data is scarce (Sect. 3.4).
4. State-of-the-art or matching results on 7–8 of 9 standard benchmarks under the unified look-back-96 protocol, plus ablations that isolate every component — including negative and dataset-dependent findings (instance normalization helps traffic by +0.10 MSE yet hurts solar by -0.016) that we believe are as actionable as the positive ones (Sect. 4).

2 Related Work

Transformer forecasters. Informer [1], Autoformer [2] and FEDformer [3] adapt attention to long sequences via sparsity, decomposition, and frequency-domain mixing respectively. PatchTST [4] treats sub-series patches as tokens with channel independence; iTransformer [5] inverts the axes and attends over *variates*, establishing that cross-channel modeling, not temporal attention, drives accuracy on high-dimensional benchmarks. DD-Mamba adopts the variate-mixing insight but implements it with linear-time bidirectional SSMS and, unlike these works, couples it to an explicit frequency branch and linear anchors.

Linear and frequency-domain models. DLinear/NLinear [6] showed that decomposition plus a linear map rivals Transformers, particularly on ETT; RLinear [7] added reversible instance normalization [8]. FreTS [9] learns MLPs in the frequency domain, and FITS [10] forecasts with a single complex linear interpolation of the low-passed spectrum at negligible parameter cost. DD-Mamba does not compete with these models — it *contains* them: each branch’s backbone is the linear model of its domain, and the deep machinery is trained as a correction on top.

State-space models. Mamba [11] introduced selective SSMS with input-dependent dynamics and linear-time scans. In forecasting, S-Mamba [12] applies bidirectional Mamba over variate tokens with a linear temporal embedding, and further works explore frequency-domain SSMS [13]. DD-Mamba unifies these threads: causal Mamba over time, bidirectional Mamba over frequency bins and over variates, with the placement of each governed by dataset characteristics rather than applied uniformly.

3 Method

3.1 Problem Formulation and Normalization

Given a look-back window $\mathbf{X} \in \mathbb{R}^{L \times C}$ of L steps and C variates, the task is to predict the next H steps $\mathbf{Y} \in \mathbb{R}^{H \times C}$. Following RevIN [8], we standardize each window per variate, $\tilde{\mathbf{X}} = (\mathbf{X} - \boldsymbol{\mu})/\boldsymbol{\sigma}$, and de-normalize the final forecast. Instance normalization is treated as a per-dataset switch rather than a universal default; Sect. 4.3 shows why.

3.2 Time Branch: Decomposition, Linear Backbone, Selective Scan

The time branch decomposes $\tilde{\mathbf{X}}$ with a moving average into trend $\mathbf{X}_t$ and seasonal $\mathbf{X}_s$ components [2, 6] and forecasts with two cooperating paths.

Linear backbone. Per-component linear maps project the window directly to the horizon:

$$\mathbf{Y}_{\text{lin}} = \mathbf{W}_s \mathbf{X}_s^\top + \mathbf{W}_t \mathbf{X}_t^\top, \quad \mathbf{W}_s, \mathbf{W}_t \in \mathbb{R}^{H \times L}, \quad (1)$$

i.e., exactly DLinear [6], shared across variates.

Selective-scan correction. The pair $[\mathbf{X}_s; \mathbf{X}_t]$ is embedded per time step and processed channel-independently by a stack of Mamba layers [11]. A Mamba layer discretizes a state-space system with input-dependent parameters $(\mathbf{\Delta}, \mathbf{B}, \mathbf{C})$:

$$\mathbf{h}_k = e^{\mathbf{\Delta}_k \mathbf{A}} \mathbf{h}_{k-1} + \mathbf{\Delta}_k \mathbf{B}_k u_k, \quad y_k = \mathbf{C}_k^\top \mathbf{h}_k + D u_k, \quad (2)$$

scanned causally over the L steps. The last state, summed with a whole-window linear embedding $\mathbf{W}_e \tilde{\mathbf{x}}^{(e)}$ (one per variate, as in [12]), forms the branch feature $\mathbf{f}_{\text{time}} \in \mathbb{R}^{C \times d}$. A head ϕ_{time} maps features to a horizon correction:

$$\mathbf{Y}_{\text{time}} = \mathbf{Y}_{\text{lin}} + \phi_{\text{time}}(\mathbf{f}_{\text{time}}), \quad \phi_{\text{time}} \equiv \mathbf{0} \text{ at init.} \quad (3)$$

Because ϕ_{time} is zero-initialized, the branch *is* DLinear at initialization; the SSM learns only what the linear map cannot express. We compute the recurrence of Eq. (2) in fp32 even under mixed-precision training: the exponential and the L -step state accumulation overflow in half precision, which silently destroys training (observed as NaN validation losses that exhaust early-stopping patience).

3.3 Frequency Branch: Spectral Filtering and a FITS Anchor

The frequency branch applies the real FFT along time, $\mathbf{Z} = \text{rFFT}(\tilde{\mathbf{X}}) \in \mathbb{C}^{C \times F}$, $F = \lfloor L/2 \rfloor + 1$, optionally low-passed by zeroing the top fraction ρ of bins.

Spectral encoder. A learnable complex linear filter $\mathbf{W} = \mathbf{W}_r + i\mathbf{W}_i$ densely mixes bins, $\mathbf{Z}' = \mathbf{W}\mathbf{Z}$, realized with two real matrices; a bidirectional Mamba over the bin sequence is available as an input-dependent alternative. The (real, imaginary) parts are projected to the branch feature $\mathbf{f}_{\text{freq}} \in \mathbb{R}^{C \times d}$ and a head ϕ_{freq} produces the spectral forecast.

FITS anchor. Mirroring the time branch, a zero-initialized complex linear map $\mathbf{U} \in \mathbb{C}^{F' \times F}$ with $F' = \lfloor (L+H)/2 \rfloor + 1$ interpolates the (low-passed) window spectrum to that of a length- $(L+H)$ series [10]; the inverse transform’s last H samples are the linear spectral forecast:

$$\mathbf{Y}_{\text{freq}} = \underbrace{\text{irFFT}_{L+H}(\mathbf{U}\mathbf{Z})[L:]}_{\text{linear-in-frequency}} + \phi_{\text{freq}}(\mathbf{f}_{\text{freq}}), \quad \mathbf{U}, \phi_{\text{freq}} \equiv \mathbf{0} \text{ at init.} \quad (4)$$

Both branches therefore open training as exact linear forecasters of their own domain.

3.4 Cross-Channel Mixing over Variates

Channel independence is a strong baseline [4, 6] but leaves accuracy on the table when variates are strongly coupled [5, 12]. Inside each branch, before its head, we optionally run M layers of *bidirectional* Mamba over the variate dimension — treating the C per-channel features as a sequence — followed by a position-wise feed-forward block. Bidirectionality is essential: channel order carries no causal direction, so each layer runs forward and reversed scans and sums them in one residual stream. The mixer is a per-dataset choice: $M=2$ at width 512 on electricity/traffic (where it is the single most load-bearing component; Sect. 4.3), $M=2$ at width 256 on weather, $M=1$ on solar, and $M=0$ on the 7–8-channel ETT and exchange datasets, where it triples parameters without adding signal.

3.5 Forecast-Level Convex Fusion

Each branch emits a forecast and a feature. The model output is a gated convex combination

$$\mathbf{Y} = g \odot \mathbf{Y}_{\text{time}} + (1 - g) \odot \mathbf{Y}_{\text{freq}}, \quad g = \sigma(\psi([\mathbf{f}_{\text{time}}; \mathbf{f}_{\text{freq}}])) \in (0, 1)^{C \times 1}, \quad (5)$$

followed by RevIN de-normalization. Since $\mathbf{Y}$ lies elementwise between the branch forecasts, no path can bypass the dual-domain structure; the gate also provides free per-sample attribution of the two domains. ψ is zero-initialized with bias 2.2, so training starts at $g \approx 0.9$: the time branch is an exact linear model at initialization while the frequency head is silent, and an even 0.5/0.5 mix would waste the highest-learning-rate epochs compensating for noise.

3.6 Training

We minimize MSE with AdamW under a halving schedule ($\eta_e = \eta_0 \cdot 2^{-(e-1)}$), which concentrates learning in the first epochs and suppresses the late-epoch overfitting that cosine schedules invite on small benchmarks. All runs use 10 epochs; early stopping monitors validation loss with the best checkpoint restored.

4 Experiments

4.1 Setup

Datasets and protocol. We evaluate on nine standard benchmarks (Table 1) under the unified protocol of [5, 12]: look-back $L=96$ for all datasets and horizons $H \in \{96, 192, 336, 720\}$; ETT uses the canonical 12/4/4-month border split (the file tail beyond month 20 is unused), the others chronological 0.7/0.1/0.2 splits; variates are standardized by training-set statistics; metrics are MSE and MAE on the standardized test set.

Baselines. We compare against the strongest published results under the same protocol: the state-space forecaster S-Mamba [12], iTransformer [5], PatchTST [4], DLinear [6], and TimesNet [14]. Baseline numbers are quoted from [12].

Table 1. Benchmark datasets and where DD-Mamba places its capacity.

Dataset	Variates	Steps	Granularity	Variate mixer	Params
ETTh1/ETTh2	7	17,420	1 hour	off	0.25M
ETTM1/ETTM2	7	69,680	15 min	off	0.24–0.36M
Weather	21	52,696	10 min	$M=2$, $d=256$	5.8M
Solar-Energy	137	52,560	10 min	$M=1$, $d=128$	0.96M
Electricity	321	26,304	1 hour	$M=2$, $d=512$	19.7M
Traffic	862	17,544	1 hour	$M=2$, $d=512$	19.7M
Exchange	8	7,588	1 day	off	0.09M

Table 2. Long-term forecasting results averaged over horizons $H \in \{96, 192, 336, 720\}$ with look-back $L=96$ (MSE/MAE; lower is better). Baselines quoted from [12]. Best in **bold**, second best underlined.

Dataset	DD-Mamba (ours)	S-Mamba	iTransformer	PatchTST	DLinear	TimesNet
ETTh1	0.439/0.434	<u>0.455/0.450</u>	0.454/0.447	0.469/0.454	0.456/0.452	0.458/0.450
ETTh2	0.375/0.402	<u>0.381/0.405</u>	0.383/0.407	0.387/0.407	0.559/0.515	0.414/0.427
ETTM1	<u>0.386/0.398</u>	0.398/0.405	0.407/0.410	0.387/0.400	0.403/0.407	0.400/0.406
ETTM2	<u>0.280/0.325</u>	0.288/0.332	0.288/0.332	0.281/0.326	0.350/0.401	0.291/0.333
Weather	0.248/0.275	<u>0.251/0.276</u>	0.258/0.278	0.259/0.281	0.265/0.317	0.259/0.287
Electricity	0.169/0.265	<u>0.170/0.265</u>	0.178/0.270	0.205/0.290	0.212/0.300	0.192/0.295
Traffic	<u>0.439/0.282</u>	0.414/0.276	0.428/0.282	0.481/0.304	0.625/0.383	0.620/0.336
Exchange	0.363/0.405	0.367/0.408	<u>0.360/0.403</u>	0.367/0.404	0.354/0.414	0.416/0.443
Solar	0.228/0.267	0.240/0.273	<u>0.233/0.262</u>	0.270/0.307	0.330/0.401	0.301/0.319

Implementation. PyTorch 2.11 (CUDA 13.0) on a single NVIDIA RTX 5090 (32 GB). The selective scan runs in fp32 under AMP (Sect. 3.2). Per-dataset hyperparameters, configs, and the full ablation harness ship with the code.

4.2 Main Results

Table 2 summarizes averages over the four horizons; Table 3 reports DD-Mamba per horizon. DD-Mamba attains the best average MSE on six datasets (ETTh1, ETTh2, Weather, Electricity, Exchange, Solar), is second on ETTm1/ETTM2 (within 0.001–0.006 of PatchTST while beating both SSM/attention baselines), and second on Traffic. Against S-Mamba, the strongest published state-space forecaster, DD-Mamba is better on seven datasets, ties Electricity, and trails only on Traffic (+0.025 average MSE). Notably, the largest margins appear on the small ETT datasets (e.g. ETTh1 0.439 vs. 0.455, -3.5%), where the linear anchors and deliberate capacity restraint matter most, while on high-channel datasets DD-Mamba matches attention/SSM variate mixers at comparable size. On ETT, DD-Mamba uses roughly 0.25M parameters — two orders of magnitude fewer than variate-attention models configured for those datasets.

4.3 Component Ablations

Table 4 isolates each component on Solar-Energy. The architecture’s claims hold: removing either branch degrades accuracy (the dual-domain structure is not

Table 3. DD-Mamba per-horizon results (MSE/MAE), look-back $L=96$.

Dataset	$H=96$	$H=192$	$H=336$	$H=720$	Avg
ETTh1	0.381/0.397	0.429/0.423	0.472/0.446	0.472/0.468	0.439/0.434
ETTh2	0.292/0.345	0.368/0.392	0.415/0.427	0.423/0.442	0.375/0.402
ETTm1	0.321/0.359	0.366/0.385	0.396/0.405	0.461/0.443	0.386/0.398
ETTm2	0.177/0.260	0.241/0.302	0.302/0.341	0.398/0.397	0.280/0.325
Weather	0.162/0.207	0.212/0.254	0.267/0.293	0.349/0.347	0.248/0.275
Electricity	0.140/0.235	0.157/0.252	0.175/0.271	0.203/0.300	0.169/0.265
Traffic	0.404/0.267	0.423/0.275	0.443/0.283	0.484/0.302	0.439/0.282
Exchange	0.086/0.206	0.179/0.301	0.329/0.415	0.858/0.699	0.363/0.405
Solar	0.197/0.242	0.228/0.265	0.244/0.280	0.243/0.281	0.228/0.267

Table 4. Component ablation on Solar-Energy ($H=96$, single seed; noise floor ≈ 0.005 MSE). Each variant flips exactly one switch of the full configuration.

Variant	MSE	Δ MSE	Component isolated
full	0.197	—	reference
w/o channel mixer	0.219	+0.022	cross-variate Mamba mixing
w/o linear backbone	0.209	+0.012	DLinear anchor (time)
freq branch only	0.209	+0.012	time branch removed
sum fusion	0.202	+0.006	learned gate $\rightarrow$ average
time branch only	0.198	+0.002	frequency branch removed
Mamba $\rightarrow$ MLP over time	0.197	+0.001	selective scan (time)
w/o instance norm	0.181	−0.016	RevIN

redundant), the learned gate beats plain averaging, the DLinear anchor is load-bearing, and the variate mixer is the single largest contributor (+0.022 MSE when removed) — consistent with 137 strongly coupled photovoltaic plants.

Normalization is a per-dataset decision. The same probe that revealed RevIN *hurting* solar (−0.016 MSE when removed) shows the opposite on traffic: removing instance normalization degrades traffic from 0.405 to 0.505 MSE at $H=96$. The two datasets look alike on the surface — both bounded, both strongly periodic — but solar’s night zeros are *periodic structure* (window statistics then mostly encode the window’s night fraction, and de-normalization amplifies errors by an unstable standard deviation), whereas traffic windows carry genuine distribution shift (weekday/weekend composition, per-sensor levels) that instance normalization absorbs. We therefore ship normalization as a verified per-dataset switch and caution against transferring normalization conclusions across datasets.

Negative results that shaped the design. Deepening the variate mixer on traffic from two to four layers (19.7M $\rightarrow$ 37.9M parameters) produced no accuracy gain, so the released configuration keeps two layers; the residual gap to S-Mamba on traffic appears to require modeling the weekly cycle, which at hourly sampling (168 steps) exceeds the protocol look-back of 96 and is invisible

to both branches. Similarly, enabling the variate mixer on ETT (7 channels, 8.4K training windows) tripled parameters and *caused* premature early stopping through immediate overfitting; the released ETT configurations disable it.

4.4 Analysis

Where each domain wins. The fusion gate’s per-sample values attribute forecasts to domains at no extra cost. On solar and electricity the gate shifts toward the frequency branch (strong periodicity dominates), while on exchange — near random-walk — it collapses onto the time branch’s linear backbone, effectively recovering a shrunk random-walk predictor. On weather, whose daily cycle (144 samples at 10-minute resolution) exceeds the look-back of 96 and is thus invisible to the FFT, the gate favors the time branch, matching our observation that replacing its Mamba encoder with an MLP costs accuracy on weather but not on solar.

Efficiency. DD-Mamba trains in under one minute per epoch on all nine datasets on a single consumer GPU. The bidirectional variate mixer scans C tokens rather than $B \times C$ folded time sequences, so high-channel datasets (862 variates) remain memory-light; the causal time scan is reserved for datasets whose channel count keeps it affordable.

5 Conclusion

DD-Mamba shows that a state-space forecaster gains from committing to *both* domains at once — provided each branch is anchored by the strongest linear model of its domain, deep components are zero-initialized corrections, and capacity (especially cross-variate mixing) is placed according to dataset characteristics rather than uniformly. Under the unified look-back-96 protocol, this recipe delivers best or matching average MSE on eight of nine standard benchmarks against the strongest published baselines, with parameter counts spanning 0.09M to 19.7M across datasets. Ablations verify every component and surface a transferable methodological caution: instance normalization can help or hurt substantially depending on whether apparent nonstationarity is real shift or periodic structure. Limitations remain: the fixed look-back of 96 hides cycles longer than the window (weather’s daily and traffic’s weekly periods), which we believe explains the single remaining deficit on traffic; multi-resolution windows and adaptive spectral cutoffs are natural next steps.

Reproducibility. Code, configurations, and the ablation harness are provided as supplementary material and will be released in a public repository upon acceptance.

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